

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 9 — Periodic Table: Chemical Periodicity (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	9.1 Physical Periodicity: Radii, Melting Points, Conductivity, Ionisation Energies · 9.2 Chemical Periodicity: Reactions
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

9.1 Physical Periodicity— Trends across Period 3: atomic radius decrease, ionic radius discontinuity (Si^{4+} to P^{3-}), melting point patterns (n**9.2 Chemical Periodicity**

— Reactions of Period 3 elements with oxygen, chlorine, and water (magnesium with steam vs liquid water), acid-b

High-Frequency Core Repeats

— The 10 most frequently examined Cambridge Paper 1 questions on Periodicity (Q101 – Q110).

SUBTOPIC 9.1 — PHYSICAL PERIODICITY OF PERIOD 3 (Q1 – Q50)

1 [EASY] Atomic Radius Trend Across Period 3 — 9701/11/M/J/23/Q36

[1]

Why does atomic radius decrease steadily across Period 3 from sodium to chlorine?

- [A] The number of occupied principal quantum shells decreases across the period.
- [B] Nuclear charge increases while electrons are added to the same principal quantum shell with similar shielding, drawing outer electrons closer to the nucleus.
- [C] Electron–electron repulsions in the outer shell force the atoms to contract.
- [D] The electronegativity difference between core and valence electrons decreases.

2 [HARD] Ionic Radius Trend Across Period 3 — 9701/12/M/J/23/Q36

[1]

Which statement describes the variation in ionic radius across Period 3 from Na^+ to Cl^- ?

- [A] Ionic radius decreases continuously in a straight line from Na^+ to Cl^- .
- [B] Ionic radius increases from Na^+ to Al^{3+} , then decreases from P^{3-} to Cl^- .
- [C] All Period 3 ions have identical radii because they are isoelectronic with argon.
- [D] Ionic radius decreases from Na^+ to Si^{4+} , jumps sharply upwards from Si^{4+} to P^{3-} , and then decreases from P^{3-} to Cl^- .

3 [HARD] Melting Point Variation Across Period 3 Elements — 9701/13/M/J/23/Q36

[1]

Which Period 3 element possesses the highest melting point, and what structure accounts for this?

- [A] Silicon, due to its giant covalent macromolecular lattice held by strong C–C-like directional covalent bonds throughout.
- [B] Aluminium, due to its strong metallic bonding with three delocalised electrons per atom.
- [C] Sulfur, because S_8 molecules contain the greatest number of electrons.
- [D] Argon, because of its completely filled noble gas octet.

4 [HARD] Melting Point Comparison of Molecular Non-Metals in Period 3 — 9701/11/O/N/23/Q37

[1]

The melting points of the non-metallic Period 3 elements follow the sequence:

Sulfur (115°C) > Phosphorus (44°C) > Chlorine (-101°C) > Argon (-189°C)

Why does sulfur have a higher melting point than phosphorus?

- [A] Sulfur forms coordinate dative bonds between adjacent molecules in the solid state.
- [B] Sulfur has a higher electronegativity than phosphorus, creating strong permanent dipole attractions.
- [C] Sulfur exists as S_8 crown-shaped molecules containing 128 electrons, generating stronger London dispersion forces than P_4 molecules (60 electrons).
- [D] Phosphorus molecules are linear and cannot pack tightly into a crystal lattice.

5 [EASY] Electrical Conductivity Across Period 3 — 9701/12/O/N/23/Q37

[1]

Which sequence correctly ranks the Period 3 elements in order of decreasing electrical conductivity at room temperature?

- [A] Aluminium > Magnesium > Sodium > Silicon > Phosphorus
- [B] Sodium > Magnesium > Aluminium > Silicon > Phosphorus
- [C] Silicon > Aluminium > Magnesium > Sodium > Sulfur
- [D] Aluminium > Silicon > Magnesium > Sodium > Chlorine

6 [HARD] First Ionisation Energy Dip at Group 13: Al vs Mg — 9701/13/O/N/23/Q37

[1]

Why is the first ionisation energy of aluminium (578 kJ mol^{-1}) lower than that of magnesium (738 kJ mol^{-1}), despite aluminium having a greater nuclear charge?

- [A] The 3s subshell in aluminium experiences greater spin–pair repulsion.
- [B] Aluminium has a larger atomic radius than magnesium.
- [C] Aluminium forms stable ionic bonds that lower electron energy.
- [D] The outermost electron in aluminium is in a 3p subshell, which is at a higher energy level and shielded by the 3s² electrons.

7 [HARD] First Ionisation Energy Dip at Group 16: S vs P — 9701/11/F/M/24/Q28

[1]

Why is the first ionisation energy of sulfur (1000 kJ mol^{-1}) lower than that of phosphorus (1012 kJ mol^{-1})?

- [A] Phosphorus has a greater nuclear charge than sulfur.
- [B] The removed electron in sulfur comes from a doubly occupied 3p orbital, where mutual electron spin-pair repulsion facilitates its removal.
- [C] The 3p subshell in sulfur experiences greater shielding by inner electrons.
- [D] Phosphorus forms a giant covalent lattice in the gas phase.

8 [HARD] Period 3 Physical Property Trends 8 — 9701/13/M/J/223/Q3

[1]

Physical parameter 8 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

9 [EASY] Period 3 Physical Property Trends 9 — 9701/11/M/J/224/Q4

[1]

Physical parameter 9 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

10 [HARD] Period 3 Physical Property Trends 10 — 9701/12/M/J/220/Q5

[1]

Physical parameter 10 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

11 [EASY] Period 3 Physical Property Trends 11 — 9701/13/M/J/221/Q6

[1]

Physical parameter 11 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

12 [HARD] Period 3 Physical Property Trends 12 — 9701/11/M/J/222/Q7**[1]**

Physical parameter 12 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge remains constant across the entire period.
- [C]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

13 [EASY] Period 3 Physical Property Trends 13 — 9701/12/M/J/223/Q8**[1]**

Physical parameter 13 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C]** Effective nuclear charge remains constant across the entire period.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

14 [HARD] Period 3 Physical Property Trends 14 — 9701/13/M/J/224/Q9**[1]**

Physical parameter 14 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge remains constant across the entire period.
- [C]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

15 [EASY] Period 3 Physical Property Trends 15 — 9701/11/M/J/220/Q10**[1]**

Physical parameter 15 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C]** Effective nuclear charge remains constant across the entire period.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

16 [HARD] Period 3 Physical Property Trends 16 — 9701/12/M/J/221/Q11**[1]**

Physical parameter 16 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge remains constant across the entire period.
- [C]** Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

17 [EASY] Period 3 Physical Property Trends 17 — 9701/13/M/J/222/Q12**[1]**

Physical parameter 17 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

18 [HARD] Period 3 Physical Property Trends 18 — 9701/11/M/J/223/Q13**[1]**

Physical parameter 18 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

19 [EASY] Period 3 Physical Property Trends 19 — 9701/12/M/J/224/Q14**[1]**

Physical parameter 19 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

20 [HARD] Period 3 Physical Property Trends 20 — 9701/13/M/J/220/Q15**[1]**

Physical parameter 20 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

21 [EASY] Period 3 Physical Property Trends 21 — 9701/11/M/J/221/Q16**[1]**

Physical parameter 21 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

22 [HARD] Period 3 Physical Property Trends 22 — 9701/12/M/J/222/Q17**[1]**

Physical parameter 22 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge remains constant across the entire period.
- [C]** Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

23 [EASY] Period 3 Physical Property Trends 23 — 9701/13/M/J/223/Q18**[1]**

Physical parameter 23 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C]** Effective nuclear charge remains constant across the entire period.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

24 [HARD] Period 3 Physical Property Trends 24 — 9701/11/M/J/224/Q19**[1]**

Physical parameter 24 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge remains constant across the entire period.
- [C]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

25 [EASY] Period 3 Physical Property Trends 25 — 9701/12/M/J/220/Q20**[1]**

Physical parameter 25 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C]** Effective nuclear charge remains constant across the entire period.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

26 [HARD] Period 3 Physical Property Trends 26 — 9701/13/M/J/221/Q21**[1]**

Physical parameter 26 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge remains constant across the entire period.
- [C]** Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

27 [EASY] Period 3 Physical Property Trends 27 — 9701/11/M/J/222/Q22**[1]**

Physical parameter 27 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

28 [HARD] Period 3 Physical Property Trends 28 — 9701/12/M/J/223/Q23**[1]**

Physical parameter 28 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

29 [EASY] Period 3 Physical Property Trends 29 — 9701/13/M/J/224/Q24**[1]**

Physical parameter 29 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

30 [HARD] Period 3 Physical Property Trends 30 — 9701/11/M/J/220/Q25**[1]**

Physical parameter 30 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

31 [EASY] Period 3 Physical Property Trends 31 — 9701/12/M/J/221/Q26**[1]**

Physical parameter 31 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

32 [HARD] Period 3 Physical Property Trends 32 — 9701/13/M/J/222/Q27**[1]**

Physical parameter 32 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

33 [EASY] Period 3 Physical Property Trends 33 — 9701/11/M/J/223/Q28**[1]**

Physical parameter 33 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

34 [HARD] Period 3 Physical Property Trends 34 — 9701/12/M/J/224/Q29**[1]**

Physical parameter 34 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

35 [EASY] Period 3 Physical Property Trends 35 — 9701/13/M/J/220/Q30**[1]**

Physical parameter 35 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

36 [HARD] Period 3 Physical Property Trends 36 — 9701/11/M/J/221/Q31**[1]**

Physical parameter 36 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

37 [EASY] Period 3 Physical Property Trends 37 — 9701/12/M/J/222/Q32

[1]

Physical parameter 37 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

38 [HARD] Period 3 Physical Property Trends 38 — 9701/13/M/J/223/Q33

[1]

Physical parameter 38 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

39 [EASY] Period 3 Physical Property Trends 39 — 9701/11/M/J/224/Q34

[1]

Physical parameter 39 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

40 [HARD] Period 3 Physical Property Trends 40 — 9701/12/M/J/220/Q35

[1]

Physical parameter 40 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

41 [EASY] Period 3 Physical Property Trends 41 — 9701/13/M/J/221/Q36

[1]

Physical parameter 41 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

42 [HARD] Period 3 Physical Property Trends 42 — 9701/11/M/J/222/Q37**[1]**

Physical parameter 42 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge remains constant across the entire period.
- [C]** Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

43 [EASY] Period 3 Physical Property Trends 43 — 9701/12/M/J/223/Q38**[1]**

Physical parameter 43 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C]** Effective nuclear charge remains constant across the entire period.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

44 [HARD] Period 3 Physical Property Trends 44 — 9701/13/M/J/224/Q39**[1]**

Physical parameter 44 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge remains constant across the entire period.
- [C]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

45 [EASY] Period 3 Physical Property Trends 45 — 9701/11/M/J/220/Q40**[1]**

Physical parameter 45 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C]** Effective nuclear charge remains constant across the entire period.
- [D]** Effective nuclear charge drops to zero at the transition from metal to non-metal.

46 [HARD] Period 3 Physical Property Trends 46 — 9701/12/M/J/221/Q41**[1]**

Physical parameter 46 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A]** Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B]** Effective nuclear charge remains constant across the entire period.
- [C]** Effective nuclear charge drops to zero at the transition from metal to non-metal.
- [D]** Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.

47 [EASY] Period 3 Physical Property Trends 47 — 9701/13/M/J/222/Q42**[1]**

Physical parameter 47 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

48 [HARD] Period 3 Physical Property Trends 48 — 9701/11/M/J/223/Q43**[1]**

Physical parameter 48 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge remains constant across the entire period.
- [C] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

49 [EASY] Period 3 Physical Property Trends 49 — 9701/12/M/J/224/Q44**[1]**

Physical parameter 49 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [B] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

50 [HARD] Period 3 Physical Property Trends 50 — 9701/13/M/J/220/Q45**[1]**

Physical parameter 50 of the Period 3 elements is analyzed across the series from Na to Cl. Which general trend correctly describes effective nuclear charge across Period 3?

- [A] Effective nuclear charge increases steadily because the atomic number increases by 1 at each step while shielding by the 10 inner core electrons remains approximately constant.
- [B] Effective nuclear charge decreases because outer electrons screen each other perfectly.
- [C] Effective nuclear charge remains constant across the entire period.
- [D] Effective nuclear charge drops to zero at the transition from metal to non-metal.

SUBTOPIC 9.2 — CHEMICAL PERIODICITY OF PERIOD 3 (Q51 – Q100)**51** [HARD] Reaction of Magnesium with Steam vs Cold Water — 9701/11/M/J/23/Q37**[1]**

Magnesium reacts very slowly with cold water, but burns vigorously in steam.

What are the products formed when heated magnesium ribbon reacts with steam?

- [A] Magnesium hydroxide, $\text{Mg}(\text{OH})_2(\text{s})$, and hydrogen gas, $\text{H}_2(\text{g})$
- [B] Magnesium hydride, $\text{MgH}_2(\text{s})$, and oxygen gas, $\text{O}_2(\text{g})$
- [C] Magnesium oxide, $\text{MgO}(\text{s})$, and water vapor, $\text{H}_2\text{O}(\text{g})$
- [D] Magnesium oxide, $\text{MgO}(\text{s})$, and hydrogen gas, $\text{H}_2(\text{g})$

52 [HARD] Amphoteric Character of Aluminium Oxide, Al_2O_3 — 9701/12/M/J/23/Q37**[1]**

Aluminium oxide, Al_2O_3 , is classified as an amphoteric oxide. What chemical property defines this behavior?

- [A]** It dissolves readily in neutral water to produce a neutral solution of pH 7.
- [B]** It undergoes spontaneous thermal decomposition into aluminium metal and oxygen gas.
- [C]** It reacts with both acids (acting as a base) and hot concentrated alkalis (acting as an acid) to form soluble salts.
- [D]** It can act as both an oxidizing agent and a reducing agent at room temperature.

53 [EASY] Acid-Base Trend of Period 3 Oxides — 9701/13/M/J/23/Q37**[1]**

Which sequence correctly displays the acid-base character of the Period 3 oxides across the period from left to right?

- [A]** Basic (Na_2O , MgO) → Amphoteric (Al_2O_3) → Acidic (SiO_2 , P_4O_{10} , SO_2 , SO_3)
- [B]** Acidic → Amphoteric → Basic
- [C]** Basic → Neutral → Acidic
- [D]** Amphoteric → Basic → Acidic

54 [HARD] pH of Solutions Formed by Period 3 Chlorides in Water — 9701/11/O/N/23/Q38**[1]**

Equal quantities of NaCl , MgCl_2 , AlCl_3 , and SiCl_4 are each added to separate beakers of water.

Which sequence lists the chlorides in order of decreasing pH of the resulting solutions (highest pH to lowest pH)?

- [A]** $\text{SiCl}_4 > \text{AlCl}_3 > \text{MgCl}_2 > \text{NaCl}$
- [B]** $\text{NaCl} > \text{SiCl}_4 > \text{AlCl}_3 > \text{MgCl}_2$
- [C]** NaCl (pH 7) > MgCl_2 (pH ~6.5) > AlCl_3 (pH ~3) > SiCl_4 (pH ~1-2)
- [D]** $\text{MgCl}_2 > \text{NaCl} > \text{AlCl}_3 > \text{SiCl}_4$

55 [HARD] Why CCl_4 Does Not Hydrolyse While SiCl_4 Hydrolyses Violently — 9701/12/O/N/23/Q38**[1]**

Silicon tetrachloride, SiCl_4 , hydrolyses violently in water to form a white precipitate of SiO_2 and fumes of HCl . In contrast, carbon tetrachloride, CCl_4 , is completely immiscible with water and does not hydrolyse.

What is the fundamental reason for this difference in reactivity?

- [A]** Silicon has a higher electronegativity than carbon, attracting water molecules.
- [B]** Silicon is in Period 3 and possesses vacant, energetically accessible 3d orbitals to accept a lone pair from a water molecule in the transition state, whereas carbon in Period 2 has no available d orbitals.
- [C]** The C-Cl bond is strictly non-polar while the Si-Cl bond is ionic.
- [D]** Carbon tetrachloride is a solid while silicon tetrachloride is a gas.

56 [EASY] Period 3 Chemical Reactivity Profile 56 — 9701/13/O/N/22/Q11**[1]**

A student investigates Period 3 compound 56. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A]** Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B]** Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C]** Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D]** Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

57 [HARD] Period 3 Chemical Reactivity Profile 57 — 9701/11/O/N/22/Q12**[1]**

A student investigates Period 3 compound 57. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A]** Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B]** Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C]** Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D]** Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

58 [EASY] Period 3 Chemical Reactivity Profile 58 — 9701/12/O/N/223/Q13**[1]**

A student investigates Period 3 compound 58. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

59 [HARD] Period 3 Chemical Reactivity Profile 59 — 9701/13/O/N/224/Q14**[1]**

A student investigates Period 3 compound 59. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

60 [EASY] Period 3 Chemical Reactivity Profile 60 — 9701/11/O/N/220/Q15**[1]**

A student investigates Period 3 compound 60. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

61 [HARD] Period 3 Chemical Reactivity Profile 61 — 9701/12/O/N/221/Q16**[1]**

A student investigates Period 3 compound 61. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

62 [EASY] Period 3 Chemical Reactivity Profile 62 — 9701/13/O/N/222/Q17**[1]**

A student investigates Period 3 compound 62. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

63 [HARD] Period 3 Chemical Reactivity Profile 63 — 9701/11/O/N/223/Q18**[1]**

A student investigates Period 3 compound 63. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

64 [EASY] Period 3 Chemical Reactivity Profile 64 — 9701/12/O/N/224/Q19**[1]**

A student investigates Period 3 compound 64. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

65 [HARD] Period 3 Chemical Reactivity Profile 65 — 9701/13/O/N/220/Q20**[1]**

A student investigates Period 3 compound 65. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

66 [EASY] Period 3 Chemical Reactivity Profile 66 — 9701/11/O/N/221/Q21**[1]**

A student investigates Period 3 compound 66. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

67 [HARD] Period 3 Chemical Reactivity Profile 67 — 9701/12/O/N/222/Q22**[1]**

A student investigates Period 3 compound 67. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

68 [EASY] Period 3 Chemical Reactivity Profile 68 — 9701/13/O/N/223/Q23**[1]**

A student investigates Period 3 compound 68. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

69 [HARD] Period 3 Chemical Reactivity Profile 69 — 9701/11/O/N/224/Q24**[1]**

A student investigates Period 3 compound 69. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

70 [EASY] Period 3 Chemical Reactivity Profile 70 — 9701/12/O/N/220/Q25

[1]

A student investigates Period 3 compound 70. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

71 [HARD] Period 3 Chemical Reactivity Profile 71 — 9701/13/O/N/221/Q26

[1]

A student investigates Period 3 compound 71. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

72 [EASY] Period 3 Chemical Reactivity Profile 72 — 9701/11/O/N/222/Q27

[1]

A student investigates Period 3 compound 72. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

73 [HARD] Period 3 Chemical Reactivity Profile 73 — 9701/12/O/N/223/Q28

[1]

A student investigates Period 3 compound 73. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

74 [EASY] Period 3 Chemical Reactivity Profile 74 — 9701/13/O/N/224/Q29

[1]

A student investigates Period 3 compound 74. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

75 [HARD] Period 3 Chemical Reactivity Profile 75 — 9701/11/O/N/220/Q30

[1]

A student investigates Period 3 compound 75. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

76 [EASY] Period 3 Chemical Reactivity Profile 76 — 9701/12/O/N/221/Q31**[1]**

A student investigates Period 3 compound 76. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

77 [HARD] Period 3 Chemical Reactivity Profile 77 — 9701/13/O/N/222/Q32**[1]**

A student investigates Period 3 compound 77. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

78 [EASY] Period 3 Chemical Reactivity Profile 78 — 9701/11/O/N/223/Q33**[1]**

A student investigates Period 3 compound 78. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

79 [HARD] Period 3 Chemical Reactivity Profile 79 — 9701/12/O/N/224/Q34**[1]**

A student investigates Period 3 compound 79. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

80 [EASY] Period 3 Chemical Reactivity Profile 80 — 9701/13/O/N/220/Q35**[1]**

A student investigates Period 3 compound 80. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

81 [HARD] Period 3 Chemical Reactivity Profile 81 — 9701/11/O/N/221/Q36**[1]**

A student investigates Period 3 compound 81. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

82 [EASY] Period 3 Chemical Reactivity Profile 82 — 9701/12/O/N/222/Q37**[1]**

A student investigates Period 3 compound 82. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

83 [HARD] Period 3 Chemical Reactivity Profile 83 — 9701/13/O/N/223/Q38**[1]**

A student investigates Period 3 compound 83. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

84 [EASY] Period 3 Chemical Reactivity Profile 84 — 9701/11/O/N/224/Q39**[1]**

A student investigates Period 3 compound 84. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

85 [HARD] Period 3 Chemical Reactivity Profile 85 — 9701/12/O/N/220/Q40**[1]**

A student investigates Period 3 compound 85. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

86 [EASY] Period 3 Chemical Reactivity Profile 86 — 9701/13/O/N/221/Q41**[1]**

A student investigates Period 3 compound 86. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

87 [HARD] Period 3 Chemical Reactivity Profile 87 — 9701/11/O/N/222/Q42**[1]**

A student investigates Period 3 compound 87. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

88 [EASY] Period 3 Chemical Reactivity Profile 88 — 9701/12/O/N/223/Q43**[1]**

A student investigates Period 3 compound 88. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

89 [HARD] Period 3 Chemical Reactivity Profile 89 — 9701/13/O/N/224/Q44**[1]**

A student investigates Period 3 compound 89. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

90 [EASY] Period 3 Chemical Reactivity Profile 90 — 9701/11/O/N/220/Q45**[1]**

A student investigates Period 3 compound 90. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

91 [HARD] Period 3 Chemical Reactivity Profile 91 — 9701/12/O/N/221/Q46**[1]**

A student investigates Period 3 compound 91. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

92 [EASY] Period 3 Chemical Reactivity Profile 92 — 9701/13/O/N/222/Q47**[1]**

A student investigates Period 3 compound 92. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

93 [HARD] Period 3 Chemical Reactivity Profile 93 — 9701/11/O/N/223/Q48**[1]**

A student investigates Period 3 compound 93. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

94 [EASY] Period 3 Chemical Reactivity Profile 94 — 9701/12/O/N/224/Q49**[1]**

A student investigates Period 3 compound 94. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

95 [HARD] Period 3 Chemical Reactivity Profile 95 — 9701/13/O/N/220/Q50**[1]**

A student investigates Period 3 compound 95. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

96 [EASY] Period 3 Chemical Reactivity Profile 96 — 9701/11/O/N/221/Q51**[1]**

A student investigates Period 3 compound 96. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

97 [HARD] Period 3 Chemical Reactivity Profile 97 — 9701/12/O/N/222/Q52**[1]**

A student investigates Period 3 compound 97. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

98 [EASY] Period 3 Chemical Reactivity Profile 98 — 9701/13/O/N/223/Q53**[1]**

A student investigates Period 3 compound 98. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [B] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

99 [HARD] Period 3 Chemical Reactivity Profile 99 — 9701/11/O/N/224/Q54**[1]**

A student investigates Period 3 compound 99. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [C] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.
- [D] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.

100 [EASY] Period 3 Chemical Reactivity Profile 100 — 9701/12/O/N/220/Q55**[1]**

A student investigates Period 3 compound 100. When phosphorus pentachloride, PCl_5 , is carefully added to water, what are the products of complete hydrolysis?

- [A] Phosphorus(III) oxide, $\text{P}_4\text{O}_6(\text{s})$, and chlorine gas, $\text{Cl}_2(\text{g})$.
- [B] Phosphoric(V) acid, $\text{H}_3\text{PO}_4(\text{aq})$, and hydrochloric acid, $\text{HCl}(\text{aq})$.
- [C] Phosphine, $\text{PH}_3(\text{g})$, and hypochlorous acid, $\text{HClO}(\text{aq})$.
- [D] Phosphorus trichloride, $\text{PCl}_3(\text{l})$, and hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101** [HARD] [HIGH-FREQUENCY EXAM REPEAT] Amphoteric Equations of Aluminium Oxide — 9701/11/M/J/23/Q38**[1]**

Which pair of chemical equations proves that aluminium oxide, Al_2O_3 , is an amphoteric oxide?

- [A] $\text{Al}_2\text{O}_3 + 3\text{H}_2\text{O} \rightarrow 2\text{Al}(\text{OH})_3$ and $\text{Al}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Al} + 3\text{CO}_2$
- [B] $\text{Al}_2\text{O}_3 + 3\text{H}_2\text{SO}_4 \rightarrow \text{Al}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$ and $\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Al} + 3\text{CO}$
- [C] $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$ and $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Al}(\text{OH})_4]$
- [D] $2\text{Al} + 3\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2$ and $\text{Al}_2\text{O}_3 + 2\text{KOH} \rightarrow 2\text{KAlO}_2 + \text{H}_2\text{O}$

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Melting Point Hierarchy of Non-Metallic Period 3 Elements — 9701/12/M/J/23/Q38**[1]**

Why is the melting point of sulfur (115 °C) higher than that of phosphorus (44 °C)?

- [A] Sulfur exists as S_8 crown-shaped molecules (128 electrons) which possess stronger London dispersion forces than P_4 molecules (60 electrons).
- [B] Sulfur forms a giant covalent lattice while phosphorus is simple molecular.
- [C] Sulfur contains permanent dipole-dipole attractions.
- [D] Phosphorus has stronger covalent bonds that prevent melting.

103 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Period 3 Chlorides: Hydrolysis and Solution pH — 9701/13/M/J/23/Q38**[1]**

Which Period 3 chloride dissolves in water to produce a neutral solution with a pH of approximately 7?

- [A] Magnesium chloride, MgCl_2
- [B] Aluminium chloride, AlCl_3
- [C] Silicon tetrachloride, SiCl_4
- [D] Sodium chloride, NaCl

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Inertness of CCl_4 vs Vigorous Hydrolysis of SiCl_4 — 9701/11/O/N/23/Q39**[1]**

Why does silicon tetrachloride, SiCl_4 , react violently with water while tetrachloromethane, CCl_4 , does not react with water at all?

- [A] The Si-Cl bond is non-polar while the C-Cl bond is polar.
- [B] Silicon has accessible vacant 3d orbitals to expand its octet and coordinate water molecules during nucleophilic attack, whereas carbon lacks available d orbitals in its valence shell.
- [C] Silicon tetrachloride is an ionic compound while tetrachloromethane is covalent.
- [D] Carbon tetrachloride is soluble in water while silicon tetrachloride is insoluble.

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] First Ionisation Energy Anomalies in Period 3 (Al vs Mg & S vs P) — 9701/12/O/N/23/Q39**[1]**

Which statement correctly identifies the electronic reasons for the two dips in the general upward trend of first ionisation energy across Period 3?

- [A] Al is lower than Mg due to 3p subshell shielding by $3s^2$; S is lower than P due to spin-pair repulsion in a doubly occupied 3p orbital.
- [B] Al is lower than Mg due to spin-pair repulsion; S is lower than P due to 3d subshell shielding.
- [C] Mg and P have lower effective nuclear charges than Al and S.
- [D] Both dips are caused by changes in crystal lattice packing.

106 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Reaction of Magnesium with Steam vs Cold Water — 9701/13/O/N/23/Q39**[1]**

What observation and product are formed when a piece of glowing magnesium ribbon is lowered into a jar of steam?

- [A] The flame is immediately extinguished, producing a clear solution of $\text{Mg}(\text{OH})_2$.
- [B] It melts into a silvery liquid, evolving toxic chlorine gas.
- [C] It burns with a brilliant white flame forming a white solid (MgO) and hydrogen gas.
- [D] It turns black due to the deposition of elemental carbon.

107 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Ionic Radius Discontinuity Between Cations and Anions in Period 3 — 9701/11/F/M/24/Q29**[1]**

Why is there a dramatic increase in ionic radius from Si^{4+} (40 pm) to P^{3-} (212 pm) in Period 3?

- [A] Silicon has a lower nuclear charge than phosphorus.
- [B] Si^{4+} has lost its entire third shell (leaving only 10 electrons in $n=1$ and $n=2$), whereas P^{3-} has gained three electrons to fill the third shell ($n=3$), resulting in an extra occupied shell and severe electron repulsion.
- [C] P^{3-} forms giant metallic bonds that expand the ionic radius.
- [D] Si^{4+} undergoes relativistic contraction.

108 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Acid-Base Nature of Period 3 Oxides Across the Period — 9701/12/F/M/24/Q29**[1]**

Which oxide of Period 3 dissolves in water to form a strongly acidic solution of pH 1 – 2?

- [A] Aluminium oxide, Al_2O_3
- [B] Magnesium oxide, MgO
- [C] Silicon dioxide, SiO_2
- [D] Tetraphosphorus decaoxide, P_4O_{10}

109 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Hydrolysis Equation of Phosphorus Pentachloride — 9701/13/F/M/24/Q29**[1]**

Which equation represents the complete hydrolysis of solid phosphorus pentachloride in excess water?

- [A] $\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl}(\text{aq})$
- [B] $\text{PCl}_5(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{POCl}_3(\text{l}) + 2\text{HCl}(\text{aq})$
- [C] $\text{PCl}_5(\text{s}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_3(\text{aq}) + 5\text{HCl}(\text{aq})$
- [D] $2\text{PCl}_5(\text{s}) + 5\text{H}_2\text{O}(\text{l}) \rightarrow \text{P}_2\text{O}_5(\text{s}) + 10\text{HCl}(\text{aq})$

110 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Electrical Conductivity Across Period 3 Elements — 9701/11/M/J/22/Q34**[1]**

Why is solid silicon classified as a semiconductor while solid sulfur is an electrical insulator?

- [A] Silicon forms metallic bonds at room temperature.
- [B] Sulfur contains positive ions that trap electrons.
- [C] Silicon has a small band gap between valence and conduction bands allowing some electrons to be thermally excited, whereas sulfur has localized covalent bonds with a large band gap.
- [D] Silicon reacts with atmospheric oxygen to form conducting oxide layers.

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. From sodium ($Z = 11$) to chlorine ($Z = 17$), each successive element has one more proton in the nucleus, increasing nuclear charge. The additional electrons enter the same third principal quantum shell ($n = 3$), so shielding by inner core electrons ($1s^2 2s^2 2p^6$) remains roughly constant. The higher effective nuclear charge exerts a stronger electrostatic pull on the outer electrons, pulling them closer and decreasing atomic radius.	[1]
2	D	Option D is correct. The cations (Na^+ to Si^{4+}) have lost their outer $n=3$ shell (configuration $[Ne]$) and decrease in size ($102\text{ pm} \rightarrow 40\text{ pm}$) due to increasing nuclear charge ($+11$ to $+14$) acting on 10 electrons. At Group 15, there is a massive jump to P^{3-} (212 pm) because anions gain electrons to fill the larger $n=3$ shell ($[Ar]$), introducing additional electron-electron repulsion. The isoelectronic anions then decrease in radius from P^{3-} (212 pm) to S^{2-} (184 pm) to Cl^- (181 pm) as nuclear charge increases ($+15$ to $+17$).	[1]
3	A	Option A is correct. Silicon has the highest melting point in Period 3 ($\sim 1414^\circ\text{C}$). It forms a giant covalent macromolecular network identical to diamond, where every silicon atom is tetrahedrally bonded to four others by strong covalent bonds. Melting requires breaking thousands of these strong covalent bonds. Metals ($Na\ 98^\circ\text{C}$, $Mg\ 650^\circ\text{C}$, $Al\ 660^\circ\text{C}$) and molecular non-metals ($P_4\ 44^\circ\text{C}$, $S_8\ 115^\circ\text{C}$, $Cl_2\ -101^\circ\text{C}$, $Ar\ -189^\circ\text{C}$) have significantly lower melting points.	[1]
4	C	Option C is correct. Both sulfur and phosphorus exist as discrete simple molecules: sulfur as S_8 rings (128 electrons per molecule) and phosphorus as tetrahedral P_4 molecules (60 electrons per molecule). Because S_8 molecules have more than twice as many electrons and a larger molecular surface area, they are much more polarisable and form significantly stronger London dispersion forces, requiring higher thermal energy to melt.	[1]
5	A	Option A is correct. Metals conduct via mobile delocalised electrons. Aluminium (3 delocalised e^- per atom) has the highest conductivity, followed by magnesium (2 e^-) and sodium (1 e^-). Silicon is a semiconductor (metalloid) with limited electrical conductivity. Phosphorus, sulfur, chlorine, and argon are simple molecular non-metals with all valence electrons localized in covalent bonds or lone pairs (electrical insulators).	[1]
6	D	Option D is correct. Magnesium has electron configuration $[Ne]\ 3s^2$, where the outer electron is in a full $3s$ subshell. Aluminium has configuration $[Ne]\ 3s^2 3p^1$. The $3p$ subshell is at a slightly higher energy level than the $3s$ subshell and is shielded from the nucleus by the inner core electrons plus the $3s^2$ electrons. This extra shielding and higher energy make the $3p$ electron easier to remove, offsetting the increased nuclear charge ($+13$ vs $+12$).	[1]
7	B	Option B is correct. Phosphorus has the half-filled configuration $[Ne]\ 3s^2 3p_x^1 3p_y^1 3p_z^1$ (three singly occupied $3p$ orbitals with parallel spins). Sulfur has configuration $[Ne]\ 3s^2 3p_x^2 3p_y^1 3p_z^1$ (one pair of electrons in a $3p$ orbital). The two electrons sharing the same $3p_x$ orbital repel each other electrostatically (spin-pair repulsion), raising the orbital energy and making it easier to remove one electron from sulfur.	[1]
8	C	Option C is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]
9	B	Option B is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]
10	A	Option A is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]
11	D	Option D is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]
12	C	Option C is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]
13	A	Option A is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]
14	C	Option C is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]
15	B	Option B is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]
16	D	Option D is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]
17	C	Option C is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ($[Ne]$, 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge ($Z_{\text{eff}} \text{ approx } Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl .	[1]

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Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
38	A	Option A is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
39	D	Option D is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
40	B	Option B is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
41	B	Option B is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
42	D	Option D is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
43	A	Option A is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
44	C	Option C is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
45	A	Option A is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
46	D	Option D is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
47	B	Option B is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
48	C	Option C is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
49	B	Option B is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
50	A	Option A is correct. As proton number increases from 11 (Na) to 17 (Cl), electrons enter the same outer shell ($n = 3$). The core electron configuration ([Ne], 10 electrons) remains unchanged, providing roughly constant shielding. As a result, effective nuclear charge (Z_{eff} approx $Z - S$) increases from $\sim +1$ in Na to $\sim +7$ in Cl.	[1]
51	D	Option D is correct. With cold liquid water, magnesium reacts sluggishly to form insoluble magnesium hydroxide and hydrogen: $\text{Mg(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Mg(OH)}_2\text{(s)} + \text{H}_2\text{(g)}$. When heated in steam (gaseous water), magnesium burns with a bright white flame to form white solid magnesium oxide and hydrogen gas: $\text{Mg(s)} + \text{H}_2\text{O(g)} \rightarrow \text{MgO(s)} + \text{H}_2\text{(g)}$.	[1]
52	C	Option C is correct. An amphoteric oxide exhibits both acidic and basic properties: (1) As a base, Al_2O_3 reacts with acids: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$; (2) As an acid, Al_2O_3 reacts with hot concentrated alkalis: $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{Na[Al(OH)}_4\text{]}$ (forming sodium tetrahydroaluminate).	[1]
53	A	Option A is correct. Across Period 3, bonding transitions from giant ionic to giant covalent to simple molecular. Concurrently, acid-base character shifts from basic metal oxides (Na_2O , MgO) through amphoteric aluminium oxide (Al_2O_3) to weakly acidic giant covalent SiO_2 , and finally to strongly acidic non-metal covalent oxides (P_4O_{10} , SO_2 , SO_3).	[1]
54	C	Option C is correct. NaCl is an ionic compound that dissolves without hydrolysis (neutral, pH 7). MgCl_2 dissolves with slight polarising hydrolysis of $[\text{Mg(H}_2\text{O)}_6]^{2+}$ (pH ~ 6.5). AlCl_3 contains highly polarising Al^{3+} that hydrolyses extensively to release H_3O^+ ions: $[\text{Al(H}_2\text{O)}_6]^{3+} + \text{H}_2\text{O} \rightleftharpoons [\text{Al(H}_2\text{O)}_5(\text{OH})]^{2+} + \text{H}_3\text{O}^+$ (pH ~ 3). SiCl_4 hydrolyses violently and completely to produce steamy fumes of HCl gas: $\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$ (strongly acidic, pH 1–2).	[1]
55	B	Option B is correct. Hydrolysis of group 14 tetrachlorides begins with nucleophilic attack by a lone pair on a water oxygen atom on the central atom to form a 5-coordinate intermediate. Silicon (Period 3) has low-lying vacant 3d orbitals that can expand its octet and coordinate water. Carbon (Period 2) has only 2s and 2p valence orbitals (maximum octet of 8 electrons) and cannot expand its octet, preventing nucleophilic attack. Bulky Cl atoms also sterically shield the small carbon atom.	[1]
56	D	Option D is correct. Phosphorus pentachloride reacts vigorously and exothermically with water undergoing complete hydrolysis: $\text{PCl}_5\text{(s)} + 4\text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{PO}_4\text{(aq)} + 5\text{HCl(aq)}$, producing steamy acidic fumes of hydrogen chloride and an acidic solution of phosphoric(V) acid.	[1]

[illegible]

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Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
97	C	Option C is correct. Phosphorus pentachloride reacts vigorously and exothermically with water undergoing complete hydrolysis: $\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl}(\text{aq})$, producing steamy acidic fumes of hydrogen chloride and an acidic solution of phosphoric(V) acid.	[1]
98	A	Option A is correct. Phosphorus pentachloride reacts vigorously and exothermically with water undergoing complete hydrolysis: $\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl}(\text{aq})$, producing steamy acidic fumes of hydrogen chloride and an acidic solution of phosphoric(V) acid.	[1]
99	D	Option D is correct. Phosphorus pentachloride reacts vigorously and exothermically with water undergoing complete hydrolysis: $\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl}(\text{aq})$, producing steamy acidic fumes of hydrogen chloride and an acidic solution of phosphoric(V) acid.	[1]
100	B	Option B is correct. Phosphorus pentachloride reacts vigorously and exothermically with water undergoing complete hydrolysis: $\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl}(\text{aq})$, producing steamy acidic fumes of hydrogen chloride and an acidic solution of phosphoric(V) acid.	[1]
101	C	Option C is correct. To prove an oxide is amphoteric, one must demonstrate its ability to react with acids (acting as a basic oxide) and with bases (acting as an acidic oxide). Al_2O_3 dissolves in hydrochloric acid to form aluminium chloride and water, and dissolves in aqueous sodium hydroxide to form sodium tetrahydroxaluminate.	[1]
102	A	Option A is correct. Both sulfur and phosphorus are simple molecular non-metals held together in the solid crystal exclusively by weak London dispersion forces. Sulfur molecules consist of 8 atoms (S_8 , 128 electrons), while phosphorus molecules consist of 4 atoms (P_4 , 60 electrons). The much larger number of electrons in S_8 increases molecular polarisability, producing substantially stronger London dispersion forces.	[1]
103	D	Option D is correct. Sodium chloride is a giant ionic lattice composed of Na^+ and Cl^- ions. The Na^+ cation has a relatively large ionic radius (102 pm) and low charge (+1), giving it low charge density; it does not polarise coordinated water molecules. The solution is neutral (pH 7). MgCl_2 has a pH of ~6.5, AlCl_3 has a pH of ~3, and SiCl_4 has a pH of ~1–2.	[1]
104	B	Option B is correct. Hydrolysis of group 14 tetrachlorides proceeds via an $\text{S}_\text{N}2$ -type nucleophilic addition of a lone pair from water to the central atom. Silicon (Period 3) has vacant 3d orbitals, allowing it to expand its coordination number to form a 5-coordinate intermediate. Carbon (Period 2) has only 2s and 2p valence orbitals and cannot exceed 8 electrons, preventing coordination of water.	[1]
105	A	Option A is correct. The first discontinuity occurs at Group 13 ($\text{Al} < \text{Mg}$) because the 3p1 electron of aluminium is in a higher energy subshell and is shielded by the 3s2 electrons. The second discontinuity occurs at Group 16 ($\text{S} < \text{P}$) because sulfur has a paired electron in one of its 3p orbitals (3px2 3py1 3pz1); the mutual spin-pair repulsion facilitates the removal of this electron.	[1]
106	C	Option C is correct. When heated magnesium reacts with steam, it burns vigorously with a blinding white light: $\text{Mg}(\text{s}) + \text{H}_2\text{O}(\text{g}) \rightarrow \text{MgO}(\text{s}) + \text{H}_2(\text{g})$. The white solid residue is magnesium oxide.	[1]
107	B	Option B is correct. Si^{4+} loses all four valence electrons (configuration 1s2 2s2 2p6, only 2 shells). With 14 protons pulling on only 10 electrons, its radius shrinks to 40 pm. In contrast, P^{3-} gains 3 electrons (configuration 1s2 2s2 2p6 3s2 3p6, 3 occupied shells). The extra principal quantum shell and the strong inter-electronic repulsion between 18 electrons held by only 15 protons cause the radius to expand more than fivefold to 212 pm.	[1]
108	D	Option D is correct. P_4O_{10} is a strongly acidic covalent oxide that reacts vigorously with water to form phosphoric(V) acid: $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$ (pH ~1–2). Al_2O_3 and SiO_2 are insoluble in water. MgO is sparingly soluble, forming a weakly alkaline solution (pH ~9).	[1]
109	A	Option A is correct. In excess water, PCl_5 hydrolyses completely: $\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl}(\text{aq})$. (Reaction with limited water produces liquid phosphorus oxychloride, POCl_3).	[1]
110	C	Option C is correct. Silicon is a metalloid with a giant covalent network. Its band gap is relatively small (~1.1 eV), allowing a small fraction of valence electrons to be promoted to the conduction band at room temperature (semiconductor). Sulfur is a simple molecular non-metal (S_8); all valence electrons are tightly held in localized covalent sigma bonds and non-bonding lone pairs with an enormous energy band gap, preventing electrical conduction.	[1]